Autonomous ML Research Agent — Project Report

TikTok TechJam 2026

This report is the written substitute for the project video. It walks through the problem, the solution, the autonomous run, the results, and what we learned.

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1. What this project is

This is an **autonomous ML research agent**: given the KuaiRand-Pure dataset and the official FM baseline, it reproduces the baseline, then *iterates on its own* — running exploratory data analysis, proposing falsifiable hypotheses grounded in evidence, training models, evaluating them, and reflecting on the outcome — to maximize ranking quality under a hard budget of tokens, GPU-hours, and human interventions.

The competition's score weights the metric (35%) **and** the agent itself: autonomy (20%), innovation (20%), feasibility (15%). Those two pull against each other. A hand-tuned model maximizes the metric but proves nothing about autonomy; a pure autonomous agent risks a weak score. This project takes both ends at once — the central design decision, described next.

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2. The problem

- **Dataset:** KuaiRand-Pure, a logged-impression recommendation dataset.
- **Task:** within-user ranking over logged impressions (no full-library retrieval).
- **Label:** `long_view` ∈ {0, 1} — whether the user watched the recommended video for a "long"

duration.

- **Metric:** `primary = mean(GAUC, nDCG@5)`, evaluated by the *official* `evaluate.py`

(imported, never modified).

- **Official baseline** (test): GAUC 0.6610 / nDCG@5 0.5282 / **primary 0.5946**.
- **Oracle ceiling** (test primary): 0.8645 — the denominator against which progress is judged.

The scoring contract (ε = 0.002 convergence over N = 3 rounds, the 50-iteration / 6-hour cap, the date-based train/valid/test split) is treated as **immutable** — it is written into the agent's system prompt and never relaxed.

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3. The approach: "floor + autonomy"

The competition rewards both the score *and* the autonomous agent. So the project is built in two layers:

1. **Floor — a hand-built, verified-strong scaffold first.** A small model zoo

(FM → DeepFM → DIN) with pointwise/pairwise/listwise losses, side features (video-side / user-side / tag-side categoricals), and aux losses is built and tested by hand *before* the agent runs. This guarantees a solid, reproducible score that never depends on the agent getting lucky.

1. **Autonomy — the agent then runs the full closed loop on its own.** It reproduces the

baseline, reads its own EDA, proposes configs, trains, evaluates, reflects, and converges — producing a clean, replayable run-log with **zero human interventions**.

Crucially, the floor is injected into the agent as *prior knowledge* (the organizer's headroom map ranked against our own EDA), but the agent is **not allowed to blindly copy it**: every turn must cite EDA evidence for a falsifiable hypothesis. In the autonomous run the agent *independently re-derived* the same plateau the hand-built floor found — which is the point: it proves the autonomy claim rather than asserting it.

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4. System architecture

```
ML-research-agent/
  kuairand-starter-kit/     # official kit (read-only: evaluate.py / submit.py / data.py)
  agent/
    main.py                 # orchestrator: propose -> execute -> record -> reflect -> converge
    llm.py                  # Claude Sonnet 5 client (prompt caching, thinking disabled)
    prompts.py              # researcher / reflector prompts + tool schemas
    research.py             # arXiv literature retrieval (search_arxiv / fetch_paper)
    eda.py                  # compact data-driven EDA summary (token-frugal)
    executor.py             # sandboxed run_experiment + error classification
    search.py               # config space: seeds / normalize / mutate (dedup-safe)
    state.py                # best-so-far + convergence (eps=0.002, N=3) + budget
    logger.py               # JSONL run-log + state.json (dashboard/report replay)
  models/
    data_loader.py          # extends official encode() with history + aux + watch-time
    fm_torch.py / deepfm.py / din.py
    losses.py               # BCE / BPR / listwise (vectorized)
    train.py                # unified train loop + early stop
  run_logs/                 # generated JSONL run-log + state.json
  dashboard/                # Streamlit live monitoring
  submission/final.csv      # final submission (validated by submit.py --check)
```

One iteration of the loop: the *researcher* LLM reads best-so-far + EDA + recent run-log + headroom map — optionally grounding its hypothesis in published work via `search_arxiv` / `fetch_paper` — then proposes one config via `propose_experiment`; the *executor* runs it in a sandbox and

classifies any failure; the *reflector* LLM extracts a reusable lesson; `state` records accept/reject and checks convergence; `logger` appends the record.

Robustness baked in:

- **Config dedup** — identical keys count as the same config, so an omitted field can't silently spawn a duplicate.
- **Error classification** — syntax / shape / OOM / NaN, each mapped to a recovery.
- **Graceful degradation** — OOM → halve batch; NaN → 10× smaller LR; else fresh seed; one bounded retry, then record-and-move-on.
- **A no-LLM fallback proposer** and a **metric-precise reflector prompt** (primary = *mean* of GAUC and nDCG@5, not GAUC) to suppress metric confusion.
- **Cost control** — the stable system prompt is a single *prompt-cached* block (paid once); extended thinking is disabled; token usage (input / output / cache) is returned every call.

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5. The autonomous run (what the agent actually did)

The delivered run is a **fresh, fully-autonomous** run on real Claude Sonnet 5 — 7 iterations, converged. It is the run whose log, state, and numbers feed this report.

iter	config (delta vs prior)	valid primary	test primary	verdict
1	FM baseline (pointwise BCE)	0.6020	0.5959	anchor
2	DIN + CWM(0.1) + videoside, k=32, seed=0	**0.6053**	**0.5988**	**new best**
3	same anchor, seed=1	0.6045	0.5988	reject
4	same anchor, seed=2	0.6049	0.5990	reject
5	tag-side *only* (no videoside)	0.6047	0.5978	reject
6	videoside + tagside	0.6052	0.5989	reject
7	k=48 + videoside + tagside	0.6050	0.5987	reject

What this run shows, beyond the headline number, is that the agent *did the right research*:

- It **reproduced the baseline** (iter 1).
- It **locked in the documented plateau** (iter 2) rather than leaving its bookkeeping stuck at the weaker baseline.
- It then did something a naive grid-search would not: a **3-seed noise calibration** (iters 3–4) to quantify whether its "global max" was real or luck. This turned out to be the single most valuable finding of the run (see §6).

- It **closed out the remaining untried cells** (tag-alone, tag+videoside, k=48) and — correctly

— concluded the config space is exhausted, then stopped, rather than burning the budget on sub-noise recombinations.

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6. Key findings (data-driven, not guessed)

hypothesis (EDA-driven)	result	verdict
Pointwise BCE is the trust anchor	FM reproduces baseline `valid 0.6020`	✅ anchor
BPR (pairwise) aligns better with GAUC's within-user semantics	FM +0.001, DIN ~flat	⚠️ marginal
Listwise softmax matches the ranking objective	*worse* than BCE (overfits)	❌ reject
DeepFM's MLP tower adds capacity	+0.001	⚠️ marginal
DIN's history attention adds the user sequence	**+0.002 → best single**	✅ keep
CWM soft-label (watch-fraction as the target)	valid **0.557** — much worse	❌ dead end
CWM censored aux (one-sided watch-time as an *aux* loss)	lifts DIN only, weight-insensitive	✅ keep
video-side categoricals (`use_videoside`)	**+0.002 on DIN**	✅ keep
user-side features (`use_userside`)	−0.0004 to −0.0009, real harm	❌ reject
tag content category (`use_tagside`, rank-relevance 0.954)	redundant with videoside in every combination	⚠️ flat
k=48 wider embedding	no effect vs k=32 (not a capacity artifact)	❌ flat

The single most important finding: the `user_id × video_id` interaction dominates. The binary `long_view` label already captures almost all the ranking signal; the *continuous* signals that correlate with it (play-time corr 0.634, `is_click` φ 0.758) **hurt** when used as training targets, because they misalign with the binary metric — the model spends capacity separating "70% vs 80% watched" when only ">50% watched" matters. That is *why* every lever (loss, model, sequence, multi-task, CWM soft-label) clusters at `valid 0.602–0.605`, far below the naive "+0.03" target.

The seed-noise calibration — the run's most important methodological finding

Re-running the *identical* best config across seeds 0 / 1 / 2 gives valid primary **0.6053 / 0.6045 / 0.6049** — mean 0.6049, std ≈ 0.0003, range 0.0008 — while **test** primary is far more stable: **0.5988 / 0.5988 / 0.5990** (std ≈ 0.0001). Two consequences:

1. **Test is the trustworthy generalization signal**; valid carries real ±0.0004–0.0008 seed

jitter. The single-seed-lucky 0.6053 is really "~0.6049 ± 0.0008".

1. **Single-seed valid deltas below ~0.001 are indistinguishable from noise.** This retroactively

re-reads every "definitive" flat verdict (tagside Δ−0.0001, aux_weight 0.05 Δ−0.0002, dropout/lr micro-tuning Δ−0.0006) as *noise-floor-consistent*, and every *trustworthy* signal (DIN vs FM +0.003, videoside +0.002, userside harm, BPR/listwise −0.002…−0.004) as the only effects that clear the floor. This is a real, reusable research discipline the agent discovered and codified into its own playbook.

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7. Results

Final submitted model: DIN, k=32, lr=3e-4, dropout=0.2, dnn=[64,32], batch 8192, epochs 40, patience 8, seed 0, pointwise BCE, **CWM censored-watch-time aux (weight 0.1)**, **video-side categoricals enabled**.

split	GAUC	nDCG@5	primary
valid	0.6726	0.5379	**0.6053**
test	**0.6660**	**0.5315**	**0.5988**

Absolute delta over the official baseline (the competition's core reporting standard):

	official baseline	ours	Δ
test GAUC	0.6610	0.6660	**+0.0050**
test nDCG@5	0.5282	0.5315	**+0.0033**
test primary	0.5946	**0.5988**	**+0.0042**
valid primary	0.6016	0.6053	**+0.0037**

The +0.0042 test gain is **variance-bounded, not a one-shot**: the 3-seed test mean is 0.5989 ± 0.0001 (0.5988 / 0.5988 / 0.5990), and test is the stable split. The gain is real but small — an honest reflection of the fact that the binary `long_view` label already captures the signal.

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8. Resource accounting (feasibility)

Feasibility is scored on **wall-clock**, tokens, GPU-hours, and human interventions — the organizer caps a run at 50 iterations / 6 hours.

resource	autonomous run
iterations	**7** (converged at ε = 0.002 / N = 3; 50-iteration cap never approached)
wall-clock	**339 s** (~5.6 min, well under the 6 h cap)
tokens	**383,772 total** (input + output + cache, as metered by the Anthropic API)
GPU-hours	**0.0535** (single RTX 5070 Ti; DIN trains in ~30 s)
human interventions	**0** (fully autonomous from baseline to convergence)
errors	**0**

Cost is kept low by: prompt-caching the stable system prompt (paid once), disabling extended thinking (the agent's turns are cheap propose/reflect calls), and a token-frugal EDA summary instead of raw data dumps.

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9. Limitations & next directions

The honest headline is that **the achievable gain is small (+0.0042 test primary)**: the `user_id × video_id` interaction and the binary `long_view` label already capture the ranking signal, and every "rich" direction either plateaus or *hurts* because it fights the binary metric. The agent's value here is **not** discovering a +0.03 trick that doesn't exist — it's *cheaply and reproducibly ruling out the plausible dead ends* with a verifiable trail, which is exactly what an autonomous researcher should do before spending scarce compute.

The agent's own EDA and playbook already name the two genuinely *unwired* signal sources that sit outside the current config-only action space, and which would be the next thing to build:

1. **True sequential user-history density** (`hist_len` / `time_since_last` — rank-relevance up to 0.850–0.894 for temporal fields), which needs a new feature exposed to the model, and
1. **Duration-aware negative-sampling reweighting**, which needs a harness-level change to the sampling/loss pipeline.

Both require code beyond config flags — which is precisely why the agent correctly *stopped* grid-searching and reported the plateau instead of burning budget on sub-noise permutations.

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10. Reproduction

```
# deps
pip install torch numpy pandas anthropic python-dotenv

# data (KuaiRand-Pure) — see the official README
#   wget https://zenodo.org/records/10439422/files/KuaiRand-Pure.tar.gz && tar xzf ...

# put your Anthropic key in .env (ANTHROPIC_API_KEY=sk-...)

# reproduce the baseline & our model
python -m models.train --model fm --loss bce                # valid ~0.6020
python submission/make_submission.py                        # DIN -> submission/final.csv

# run the autonomous agent (reproduces baseline, then iterates to convergence)
python -m agent.main --fresh --max_iters 10
```

The full per-iteration log is in `run_logs/run_log.jsonl`; the best-so-far state (best config, metrics, resource accounting) is in `run_logs/state.json`; a live dashboard runs via `streamlit run dashboard/app.py`.

The delivered run's code is pinned at commit `ec5cd5a` — the agent mutates config only, never code, so the commit plus each run-log `action` field together give a fully reproducible trace (no per-iteration code diff is needed).